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Cleaning sheath for power tools with cutting blades and methods of using

Abstract

A cleaning sheath for power tools, including a sheath assembly, wherein the sheath assembly includes a sheath, and wherein the sheath is rigid, the sheath is made of an impermeable material, and the sheath is hollow, a lid assembly located at one end of the sheath, a loop assembly located adjacent to the lid assembly, a cleaning solution located within a portion of the sheath, wherein the cleaning solution is adapted to clean and disinfect power tool blades located within the sheath, and an attaching assembly, wherein the attaching assembly includes an attaching element, and wherein the attaching element is removably attached to the sheath, and the attaching element is removably attached to predetermined surfaces.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3042087	12/1961	Otoupalik	N/A	N/A
5489051	12/1995	Robinson	220/570	A45F 5/02
6219921	12/2000	Pettersson	N/A	N/A
8783538	12/2013	Dattilo	224/679	A45F 5/021
8887940	12/2013	Kiceniuk, Jr.	248/339	B44D 3/127
2019/0090716	12/2018	Martinez	N/A	A47L 1/06
2021/0059390	12/2020	Knoblauch	N/A	B25H 3/022
2022/0304458	12/2021	Davis	N/A	B08B 1/32
2024/0108120	12/2023	Knoblauch	N/A	A45F 5/021

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2007141775	12/2006	WO	A47L 13/58

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part of U.S. patent application Ser. No. 17/873,376, filed on Jul. 26, 2022, the disclosure of which is hereby incorporated by reference in its entirety to provide continuity of disclosure to the extent such a disclosure is not inconsistent with the disclosure herein.

BACKGROUND OF THE INVENTION
Field of the Invention
(1) The present invention relates to a cleaning sheath for power tools with cutting blades and, more particularly, to a cleaning sheath for power tools with cutting blades that allows to easily clean the blades after utilizing the power tool through the use of mineral spirits.
Description of the Related Art
(2) Several designs for cleaners for power tools have been designed in the past. None of them, however, include a sheath where a power tool with cutting blades can be cleaned.
(3) The applicant believes that a related reference corresponds to U.S. Pat. No. 3,042,087 issued for

a protective sheath for the blade of a chainsaw. The applicant believes that another related reference corresponds to U.S. Pat. No. 6,219,921 issued for a chainsaw guide bar with an internal channel that is configured to receive a cleaning liquid. None of these references, however, teach or suggest a cleaning apparatus for power tools with rotating or reciprocating blades such as chainsaws or hedge trimmers that is comprised of a rigid, liquid impermeable sheath which is filled with a cleaning solution and configured to receive the blade of the tool such that the blade can be soaked and agitated in the cleaning solution. Furthermore, none of these references teach or suggest the providing the cleaning apparatus for power tools with rotating or reciprocating blades or guide bars with a splash resistant lid, a drain located at the bottom of the cleaning apparatus, a window or other similar viewing device on the cleaning apparatus that allows the user to see level of the cleaning solution in the cleaning apparatus, and a stand for retaining the cleaning apparatus in an upright position.

(4) Other documents describing the closest subject matter provide for a number of more or less complicated features that fail to solve the problem in an efficient and economical way. None of these patents suggest the novel features of the present invention.

SUMMARY OF THE INVENTION

(5) It is one of the objects of the present invention to provide a cleaning sheath for power tools that includes fasteners to fix the cleaning sheath to any suitable surface.

(6) It is another object of this invention to provide a cleaning sheath for power tools that includes an attaching element that allows to removably attach the cleaning sheath to working benches or any other suitable surface.

(7) It is still another object of the present invention to provide a cleaning sheath for power tools that includes an impermeable sheath to store cleaning liquid therein.

(8) It is still another object of the present invention to provide a cleaning sheath that includes a splash resistant lid.

(9) It is still another object of the present invention to provide a cleaning sheath that includes a drain located at the bottom of the cleaning apparatus for assisting in removing the cleaning fluid from the cleaning sheath.

(10) It is still another object of the present invention to provide a cleaning sheath that includes a window or other similar viewing device on the cleaning apparatus that allows the user to see level of the cleaning solution in the cleaning apparatus.

(11) It is still another object of the present invention to provide a cleaning sheath that includes a stand for retaining the cleaning apparatus in an upright position.

(12) It is yet another object of this invention to provide such a device that is inexpensive to implement and maintain while retaining its effectiveness.

(13) Further objects of the invention will be brought out in the following part of the specification, wherein detailed description is for the purpose of fully disclosing the invention without placing limitations thereon.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) With the above and other related objects in view, the invention consists in the details of construction and combination of parts as will be more fully understood from the following description, when read in conjunction with the accompanying drawings in which:

(2) FIG. **1** represents an isometric operational view of the present invention **10** fixed to a working bench.

(3) FIG. **2** shows an isometric view of the present invention **10**. The present invention includes a sheath assembly **20** and an attaching assembly **40**.

(4) FIG. **3** illustrates a rear view of the present invention **10**.
(5) FIG. **4** is a representation of a left side view of the present invention **10**.
(6) FIG. **5** is a schematic illustration of the lid assembly **30** of the present invention **10**.
(7) FIG. **6** is a schematic illustration of the present invention **10** being retained in a stand assembly **50**.

DETAILED DESCRIPTION OF THE EMBODIMENTS OF THE INVENTION

(8) Referring now to the drawings, where the present invention is generally referred to with numeral **10**, it can be observed that it basically includes power tool **12** having an elongated tool extension **14** such as a guide bar, sheath assembly **20**, a lid assembly **30**, an attaching assembly **40**, and a stand assembly **50**. It should be understood there are modifications and variations of the invention that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.

(9) As shown in FIGS. **2-4**, the sheath assembly **20** may include a sheath **22**. In a preferred embodiment the sheath **22** may be made of metal. It also may be suitable for the sheath **22** to be made of plastic, ceramic, wood, leather, or any other suitable material. The sheath **22** may have an anti-rust layer located on the exterior and interior of the sheath. The sheath **22** may substantially have a rectangular shape with rounded tip. It also may be suitable for the sheath **22** to have a triangular shape, a curved shape, a circular shape, or any other suitable shape. The shape of the sheath **22** may preferably adjust to the shape of power tools with rotating or reciprocating blades. An upper portion of the sheath **22** may be bigger in length than the lower portion of the sheath **22** to allow an easier introduction of power tools with rotating or reciprocating blades and avoiding the spill of liquids. The sheath **22** may be hollow. The uppermost end of the sheath **22** may have an opening **24**. The opening **24** of the uppermost end of the sheath **22** may have an extension that allows predetermined sized power tools with rotating or reciprocating blades or guide bars to fit thereinto.

(10) Cleaning solutions such as, but not limited to, mineral spirits may be deposited into the sheath **22** to clean power tools with rotating or reciprocating blades. The mineral spirits may also be used to disinfect the power tools with rotating or reciprocating blades or guide bars. In a preferred embodiment the power tools are hedge trimmers. It should be understood that the power tools with rotating or reciprocating blades or guide bars **14** may also include but are not limited to smaller chainsaws and band pruners. In a preferred embodiment, a power tool with rotating or reciprocating blades may be inserted and actuated into the amount **26** of solution of mineral spirits inside the sheath **22** to be cleaned. It also may be suitable to place different blade tools or any metal tools to be cleaned into the sheath **22**. Blade tools include but are not limited to hand shears, machetes, and loppers. It also may be suitable to use the sheath **22** for leaving the power tools or any metal tool into the amount **23** of cleaning fluid (e.g., mineral spirits) to be cleaned overnight. The sheath **22** may also be used to store and protect blades or guide bars **14**, power tools with rotating or reciprocating blades or swords. The sheath **22** may be attached or fixed to a predetermined surface by means of the attaching assembly **40**.

(11) Referring now to the FIG. **3** it can be observed that the attaching assembly **40** may include an attaching element **42**, junction elements **44**, openings **46** and at least one fixing element **48**. The attaching element **42** may have a front wall **41**, a top wall **45**, and a rear wall **40**. The front wall **41** of the attaching element **42** may be planar. The front wall **41** of the attaching element **42** may be removably attached to the sheath **22** through the junction elements **44**. The junction elements **44** may be nuts and bolts, screws, or any other suitable element that allows joining the attaching element **42** to the back of the sheath **22**. The front wall **41** may be attached to the top wall **45** of the attaching element **42**. The top wall **45** of the attaching element **42** may have a predetermined length. The predetermined length of the top wall **45** of the attaching element may depend on the

extension of a predetermined surface of a structure **43** where the sheath **22** is being attached. The rear wall **40** of the attaching element **42** may include openings **46**. The attaching element **42** may be used to removably attach the sheath **22** to predetermined structures such as a work bench. The top wall **45** and the rear wall **40** of the attaching element **42** may be planar. The rear wall **40** of the attaching element **42** may be a bent portion of the top wall **45** of the attaching element **42**. The attaching element **42** may be made of metal, plastic, ceramic, wood, leather, or any other suitable material.

(12) The at least one fixing element **48** may be used to fix the attaching element to predetermined surfaces. The at least one fixing element **48** may go through the openings **46**. The at least one fixing element **48** may be inserted into a surface. The openings **46** may transversely go through the rear wall of the attaching element **42**. The at least one fixing element **48** may be nuts and bolts, screws, hook and loop fasteners, suction cups, or any other suitable fixing element. The at least one fixing element **48** may be used to prevent movement of the sheath **22** when a power tool with rotating or reciprocating blades or guide bars **14** is inserted and actuated into the sheath **22**.

(13) A unique aspect of the present invention is that sheath **22** can also include a splash resistant lid assembly **30**. As shown in FIGS. **1-5**, lid assembly **30** includes, in part, lid **31**, lid hinge **32**, lid gasket **33**, lid retainer **34**, lid extension **36**, opening **38**, and loop assembly **100**. Preferably, lid **31**, lid hinge **32**, lid gasket **33**, lid retainer **34**, and lid extension **36** are constructed of any suitable, durable, UV resistant, high strength, chemical resistant polymeric material. Preferably, loop assembly **100** is constructed of any suitable, durable, UV resistant, high strength, chemical resistant material.

(14) In one embodiment, opening **38** is formed in lid retainer **34** by conventional forming techniques such as molding, stamping, forming, or the like. As discussed above, opening **38** is formed in lid retainer **34** so that the cleaning solution is not able to easily splash out of or otherwise able to escape sheath **22** while allowing the rotating or reciprocating blades or guide bars **14** to be inserted into sheath **22**. Furthermore, opening **38** is located in lid retainer **34** so that when the power tool **12** is extracted or otherwise removed from sheath **22**, the sides of opening **38** will interact with the sides of the rotating or reciprocating blades or guide bars **14** of the power tool **12** in order to remove an excess cleaning solution located on the rotating or reciprocating blades or guide bars **14** of the power tool **12**.

(15) Another unique aspect of the present invention is that sheath **22** can also include a drain **26** located at the bottom of the cleaning apparatus for assisting in removing the cleaning fluid **23** from the sheath **22**. In one embodiment, the drain **26** can be a stop cock style drain, a plug style drain, or any suitable style of drain as long as the drain **26** will allow the cleaning fluid **23** to properly drain from the sheath **22**.

(16) A still another unique aspect of the present invention is that sheath **22** includes a window **21** or other similar viewing device that allows the user to see level of the cleaning fluid **23** in the sheath **22**. The window **21** can be attached to the sheath **22** by conventional attachment techniques such as adhesives or the like. In one embodiment, the window **21** should be constructed of any suitable, durable, UV resistant, chemical resistant, transparent polymeric material. In another embodiment, the window **21** can include a series of markings or graduations **25** that will assist the user is determining the amount of cleaning fluid **23** in the sheath **22**.

(17) A still another unique aspect of the present invention is that sheath **22** includes a stand assembly **50** for retaining the cleaning apparatus in an upright position. As shown in FIG. **6**, in one embodiment, the stand assembly **50** includes a plurality of walls **52** and stand **54** and an opening **56**. Preferably, walls **52** and stand **54** are constructed of any suitable, durable, high-strength, UV resistant, chemical resistant material. Also, the walls **52** and stand **54** are secured together by conventional techniques such as fasteners, adhesives or the like. The opening **56** is formed when walls **52** are conventionally attached together by conventional techniques such as adhesives, fasteners, or the like. As shown in FIG. **6**, sheath **22** is retained in opening **56** of stand assembly **50**.

In this manner, sheath **22** can be easily stored in an upright position.
(18) A still further unique aspect of the present invention is loop assembly **100**. As shown in FIGS. **1-3**, **5** and **6**, loop assembly **100** is conventionally attached to sheath **22** at a location that is adjacent to the lid **31**. In this manner, loop assembly **100** can be used to attach the sheath **22** to a conventional utility belt (not shown) so that the sheath **22** can be carried by the user in a hand-free manner.
(19) The foregoing description conveys the best understanding of the objectives and advantages of the present invention. Different embodiments may be made of the inventive concept of this invention. It is to be understood that all matter disclosed herein is to be interpreted merely as illustrative, and not in a limiting sense.

Claims

1. A cleaning and disinfecting system, comprising: a sheath, wherein the sheath is rigid, wherein the sheath is made of an impermeable material, and wherein the sheath is hollow; a cleaning and disinfecting solution located within a portion of the sheath, and wherein the cleaning and disinfecting solution includes mineral spirits; a power tool having power tool blades, wherein the power tool blades are retained within a portion of the sheath, and wherein the power tool blades interact with the cleaning and disinfecting solution to clean and disinfect the power tool blades; a lid assembly located at one end of the sheath, wherein the lid assembly is further comprised of; a lid, a lid hinge rotatably connected to the lid and the one end of the sheath, a lid retainer, wherein a first opening is located in the lid retainer, wherein sides of the first opening are configured to interact with sides of the power tool blades in order to remove an excess of the cleaning and disinfecting solution located on sides of the power tool blades, a lid gasket located on a first side of the lid, and a lid extension located on one edge of the lid; and an attaching assembly, wherein the attaching assembly includes an attaching element such that the attaching element is removably attached to the sheath, and the attaching element is configured to be able to be removably attached to a structure.
2. The cleaning and disinfecting system, as in claim 1, wherein the attaching element is further comprised of: at least one first fastener, wherein the at least one first fastener is configured to allow the attaching element to be removably attached to the sheath.
3. The cleaning and disinfecting system, as in claim 1, wherein the sheath is further comprised of: a second opening, wherein the second opening is configured to fit the power tool blades thereinto.
4. The cleaning and disinfecting system, as in claim 1, wherein the attaching element is further comprised of: a first wall; a second wall connected to the first wall; and a third front wall connected to the first wall and located predetermined distance away from the second wall.
5. The cleaning and disinfecting system, as in claim 4, wherein the third wall is removably attached to the sheath.
6. The cleaning and disinfecting system, as in claim 4, wherein the second wall has a plurality of third openings.
7. The cleaning and disinfecting system, as in claim 6, wherein the attaching assembly is further comprised of: at least one second fastener, wherein the at least one second fastener is located within the at least one of the plurality of third openings to attach the sheath to the structure.
8. The cleaning and disinfecting system, as in claim 1, wherein the power tool blades are further comprised of: hedge trimmer blades.
9. The cleaning and disinfecting system, as in claim 1, wherein the sheath assembly is further comprised of: a loop assembly connected to the sheath adjacent to the lid assembly.
10. The cleaning and disinfecting system, as in claim 1, wherein the sheath assembly is further comprised of: a window located along a portion of the sheath; and a plurality of markings located on the window.

11. The cleaning and disinfecting system, as in claim 1, wherein the cleaning sheath is further comprised of: a stand for holding the sheath in an upright position.